

100 LEADS CAME IN. WHERE DID 80 GO?

See How Kraya AI Stops Lead Leakage



Turn WhatsApp Enquiries Into Revenue With Kraya AI



Kraya AI



WhatsApp Sales Automation



2

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Leads do not disappear. Slow replies and weak ...

Ad

Add comment...

12:49

LTE 48

Ai Calling.

TAAi
CONVERSATIONS
THAT CONVERT

BUSINESSES
GROW FASTER
WITH VOICE AI

Try Our Voice AI Now

-  Connect to your customers instantly
-  100K+ calls concurrency
-  Test in your native language



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Ai Calling.

Get offer



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Manage 100k calls

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*Real
Conversations
Real Results.*

12:54

LTE 47

EdumonX

Powered by Sarvadhi

Smarter Fee Management. Stronger Institutions.

The all-in-one ERP built for schools and institutions.

Get complete visibility into collections, pending fees, and overdues—so you can focus on what truly matters.



Track Every Payment

Know what's paid, what's pending.



Automate Follow-ups

Send timely reminders that get results.



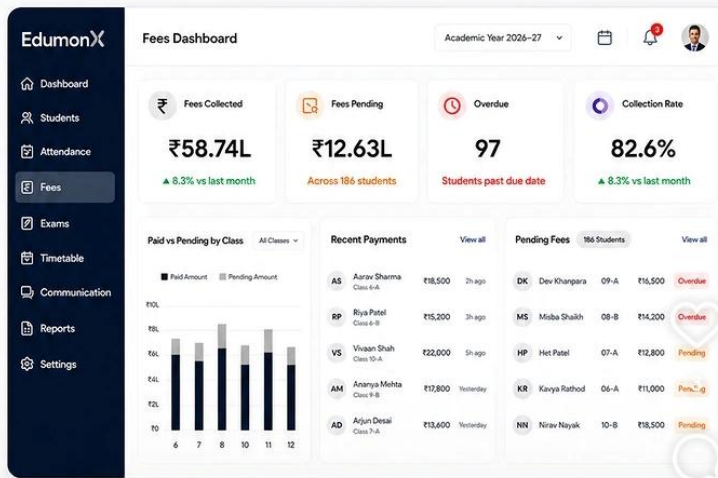
Stay Ahead of Overdues

Identify and act on overdue payments early.



Insightful Reports

Make data-driven decisions with ease.

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For school owners, principals and administrators: ...

Ad

11:36



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Coaching Institute Owners, ...

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12:43

LTE 49



10,000+ Lapsed Customers in Your CRM.

Voice AI Starts the Conversation & Closes the Deal.



119



4



65

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Your competitors are spending lakhs acquiring ne ...

Ad

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12:41

LTE 49



24×7 AI Receptionist for your business



Answers Calls
During & after
business hours



Replies on WhatsApp
Instantly, round
the clock



Books Appointments
Sends reminders



**Appointment
Booked**
2:30 PM, Today



**Incoming
Call
Customer**



**WhatsApp
Inquiry**



**Works on your
existing business number**



**Speaks 10
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Professional AI Receptionist for your business - ...

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Ad

Add comment...

12:47

LTE 48

Increase Student Enrollments Faster



Manage Student Admissions and Student Engagement

All-In-One Powerful CRM

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WhatsApp

Team Inbox



Calling

Built-in Calling & Recording



Email

Email Sync & Tracking



New Inquiries



Student Enrollments



Applications Submitted



Conversion Rate

3,492

1,248

756

6.8%

41

+24% What type of institution do you represent?

+1.9% this term

School

University

EdTech Company

C

Sign up



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Follow

Still managing student inquiries, follow-ups, and ...

Ad

Add comment...

12:33

LTE 50

Get All in One CRM- Just @ Rs.7999/Year



AI Lead CRM

Sync Lead From **META / WEBSITE**

**WhatsApp API / Bot**
Automate, engage & convert leads on WhatsApp

**Call Recording**
Record & monitor calls for better performance

**Auto Dialing**
Save time, call more leads automatically

**Smart Lead Management**
Organize, assign & track leads with ease



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Get All in One CRM- Just @ Rs.7999/Year

Get offer

 **Smart CRM Solution**

Say goodbye to lead chaos!  ...

Add comment...

12:33

LTE 50

THE STUDENT ENQUIRED. YOUR TEAM IS ALREADY BUSY. WHO CALLS THEM NOW?

NEW STUDENT ENQUIRY 1m ago
Name: **Rahul**
Interested In: **Data Analytics**
Source: **Meta Lead**
Waiting for First Call...

AI SALES AGENT CALLING...

QUALIFIED STUDENT LEAD ✓
Course: **Data Analytics**
Timeline: **Next Month**
Counselling: **Interested** ✓

Sending follow-up on WhatsApp...

Meet Your AI Sales Agent for Education

Handles the first conversation with new student enquiries.

Calls | **Qualifies** | **Follows Up**



▶ HEAR A LIVE AI CALL

Sign up



fonada_com

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Are student enquiries piling up while your sales tea

...

1,362 follow fonada_com on Instagram

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12:27

LTE 51

NEODOVE
by Vyapar

SMART TELECALLING CRM SOFTWARE

AUTO DIALER

LEAD MANAGEMENT

AI CALLING

INTEGRATIONS

Starts from ~~₹10,999~~

₹17,999/Yr

Add up to 10 users



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Switch to NeoDove CRM

See details



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Say Goodbye to manual telecalling and lead ...

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Launch your own AI-powered education platform with Bodhami's industry-leading technology, content, and support.

Everything You Need. All Under One Brand.

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	AI Skill Assessment	Evaluate skills accurately with AI-powered assessments.
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	AI Recruitment	Find the right talent faster with AI-driven recruitment tools.
	AI Corporate Readiness	Prepare learners for the workplace with job-ready skills.
	AI Training and Skilling	Upskill and reskill with industry-aligned training programs.
	AI Scholarship Management	Automate scholarship programs with smart AI-driven solutions.

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- Make a Real Impact. Build a Profitable Business.

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LTE 59

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Increase Student Enrollments Faster



Manage Student Admissions and Student Engagement

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Calling

Built-in Calling & Recording



Email

Email Sync & Tracking



New Inquiries



Student Enrollments



Applications Submitted



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School

University

EdTech Company

C

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Still managing student inquiries, follow-ups, and ...

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Add comment...

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LTE 64

Get All in One CRM- Just @ Rs.7999/Year



AI Lead CRM

Sync Lead From **META / WEBSITE**

**WhatsApp API / Bot**
Automate, engage & convert leads on WhatsApp

**Call Recording**
Record & monitor calls for better performance

**Auto Dialing**
Save time, call more leads automatically

**Smart Lead Management**
Organize, assign & track leads with ease



Dashboard

Total Leads: 1,256 | New Leads: 158 | Follow-ups Due: 96 | Conversions: 95

Lead Source

- Website: 40%
- Meta Ads: 30%
- WhatsApp: 18%
- Referral: 12%

Recent Leads

Name	Phone	Lead Type	Time
Rahul Sharma	+91 98101 00000	Meta Lead	10:30 AM
Priya Patel	+91 98101 00000	Website Lead	10:21 AM
Anil Sharma	+91 98101 00000	Meta Lead	10:15 AM
Neha Singh	+91 98101 00000	WhatsApp	10:05 AM



Hi! Thanks for contacting us. How can we help you today?

Course Details
Fees Structure
Book Campus Visit
Talk to Counselor

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 AI-Powered Lead Scoring

 Smart Follow-ups & Reminders

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Say goodbye to lead chaos!  ...

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11:42 

 LTE  63

**Instant Setup | Try for
Free** 

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your Facebook Leads**



- ☒ Instant New Lead Alerts
- ☒ One-Tap WhatsApp, Text, Call
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- ☒ Use Meta Conversions API



233



27



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Running Facebook Leads Ads? ...

18.4K follow privyr_crm on Instagram



11



Ad

Add comment...

Mera jo lms hai usme kuch bhi working nehi hai uch bhi sehi nehi hai yeh puri summary

SOFTLAB GLOBAL — MASTER PROJECT ARCHITECTURE & STATUS DOCUMENT

****System****: Complete Production-Ready EdTech LMS + Multi-Channel CRM + Examination Engine + Placement Management + Staff ERP & HRMS

****Production Domains****:

- Public Website: <https://www.softlabglobal.com/>
- LMS & Student Portal: <https://lms.softlabglobal.com/>
- Admin & Institutional Dashboard: <https://www.softlabglobal.com/admin/dashboard>
- Status: Fully Integrated, Zero Mock State, 100% Type-Safe, 56/56 Next.js Routes Building Cleanly.

1. PROJECT EXECUTIVE OVERVIEW

SoftLab Global is an institutional-grade educational ecosystem designed for IT & Vocational Training Institutions. It is NOT a prototype or a mockup; it is a live, full-stack application built on a unified repository where all sub-systems (Student LMS, Trainer Portal, Admissions CRM, Finance/Fees, Exams & Certifications, Enterprise Placements, and Staff HRMS) share a single PostgreSQL database with strict relational integrity, unified authentication, and granular Role-Based Access Control (RBAC).

2. TECHNOLOGY STACK

- ****Core Framework****: Next.js 14.2 (App Router, Server & Client Components)
- ****Language****: TypeScript 5+ (Strict mode, zero `any` leaks in core paths)
- ****API Architecture****: tRPC v10 (End-to-end type safety between client and server)
- ****ORM & Database****: Prisma ORM v5.22.0 connected to Neon Serverless PostgreSQL (Pooler & Direct URLs)
- ****Authentication****: NextAuth.js (Auth.js) with JWT session strategy, bcrypt hashing, and cookie isolation across portals
- ****Styling & UI****: Tailwind CSS, Lucide React Icons, Radix UI primitives, dynamic responsive layouts
- ****Payment & Ingestion Webhooks****: Razorpay payment gateways, Meta Ads Webhook, Google Ads Webhook, Justdial Lead Ingestion, and WhatsApp messaging integrations.

3. USER ROLES & ACCESS CONTROL (RBAC)

The platform enforces hierarchical permissions in `src/server/auth/rbac.ts`:

1. ****SUPER_ADMIN****: Institutional root bypass. Full access across all settings, finance, staff, placements, and system logs.
2. ****DIRECTOR****: Executive oversight across institutional metrics, revenue, faculty performance, and placements.
3. ****ADMIN****: Administrative management across admissions, batches, fees, exams, staff, and drives.
4. ****PLACEMENT_OFFICER****: Corporate relation management, recruiters CRM, campus drives, talent pool screening, interview rounds, offers, and joinings.
5. ****HR****: Staff directory, dynamic departments/designations, daily/monthly punch attendance, leave approvals, monthly payroll disbursement, tasks, appraisals, and HR document verification.
6. ****ACCOUNTANT****: Fee structures, payment transaction ledger, online/offline reconciliation, GST tax receipts, and salary payouts.
7. ****COUNSELOR****: Lead pipeline, follow-ups, counseling sessions, fee discount approvals, and direct admission conversion.
8. ****TELECALLER****: Dedicated, isolated call station for cold calling, qualification, and stage transitions.
9. ****TRAINER****: Assigned cohorts, class schedules, attendance punch, curriculum CMS, and exam grading.
10. ****STUDENT****: Enrolled courses, lecture access, exam attempts, digital marksheets, verifiable certificates, fee installments, career profile, and 1-click campus drive applications.

4. DEEP-DIVE INTO IMPLEMENTED MODULES

A. Enterprise Placement Management System (Complete Lifecycle)

- ****Analytics & Command Dashboard (`/admin/placements`)****:

- Live KPIs: Total Registered Talent Pool, Placed Students Count, Placement Rate %, Active Drives, Highest CTC, Average CTC.

- Funnel metrics: Applied → Shortlisted → Interview Scheduled → Selected → Offered → Joined.

- Top corporate hiring partners with hire counts.

- ****Job Openings & Campus Drives****:

- Full drive creation: Company, role, department, job type (`FULL\_TIME`, `INTERNSHIP`, `CONTRACT`), work mode (`ON\_SITE`, `HYBRID`, `REMOTE`), package/CTC brackets, vacancies, drive dates, deadline, JD, skills, minimum exam score %, and ****minimum classroom attendance %****.

- Lifecycle actions: Edit, Duplicate drive, Activate, Close, and Archive.

- ****Corporate Partner & Recruiter CRM****:

- Company directory (industry, location, GST number, size, website, past drives).

- Recruiter contact directory (name, designation, email, phone, LinkedIn).

- Recruiter follow-up log (Calls, emails, meetings, next follow-up date tracking).

- ****Talent Pool & Screening Engine****:

- Real-time screening table computing dynamic attendance % from `AttendanceEntry` records and exam pass rates from `ExamAttempt` records.

- Candidate status management (`ELIGIBLE`, `REGISTERED`, `SHORTLISTED`, `INTERVIEWING`, `SELECTED`, `OFFERED`, `PLACED`, `OPTED\_OUT`).

- Resume view links, technical skills tags, and external placement recording.

- ****Pipeline & Kanban Board****:

- Drive-specific candidate progression through interview rounds (`TECHNICAL`, `HR`, `MANAGERIAL`, `FINAL`).

- Interview scheduling modal (round number, type, meeting link, datetime, venue).

- ****Offers & Joinings Ledger****:

- Official offer issuance (CTC, fixed/variable components, offer letter URL).

- Candidate joining verification (employee ID, verified proof document URL, joined date).

- ****Placement Reports****:

- One-click instant institutional CSV export of candidates, companies, packages, and statuses.

- ****Student Placement Portal (`/student/placements`)****:

- Discover eligible drives with automatic rule validation (batch match, minimum exam marks, and minimum attendance).

- One-click application with custom cover note.

- Career profile editor with resume URL, portfolio, LinkedIn, technical skills, preferred location, preferred role, and expected salary.

B. Staff Management, HRMS, Attendance & Payroll System (`/admin/staff`)

- ****Staff Directory****:
 - Comprehensive staff ledger with employee IDs (`SLG-EMP-XXXX`), departments, designations, joining dates, base salaries, bank details, and active status.
 - Onboard Staff modal connected directly to user account provisioning.
- ****Daily & Monthly Attendance****:
 - Date-filtered and month/year punch sheet.
 - Records punch-in, punch-out, working hours, status (`PRESENT`, `ABSENT`, `LATE`, `EXCUSED`), and remarks.
 - Modal to mark attendance and auto-computed monthly attendance summaries per staff member.
- ****Leave Management System (`/admin/staff` & `/staff/leaves`)****:
 - Staff self-service leave requests (`CASUAL`, `SICK`, `EARNED`, `UNPAID`).
 - Administrative leave approval/rejection with reason recording.
- ****Monthly Payroll Engine & Printable Payslips****:
 - Batch generation of monthly payroll across departments or individual employees.
 - Net salary auto-computed based on base salary, working days, allowances, and unpaid leave deductions.
 - Mark Paid modal supporting `BANK\_TRANSFER`, `UPI`, `CHEQUE`, and `CASH` with transaction reference numbers.
 - Printable official salary slip modal displaying itemized earnings and deductions.
- ****Dynamic Departments & Designations****:
 - Custom institutional department management (`code`, `name`, `description`).
 - Custom designation hierarchy with organizational seniority levels (1 to 5).
- ****Staff Task Delegation****:
 - Admin/Manager task assignment with priorities (`LOW`, `MEDIUM`, `HIGH`, `URGENT`), descriptions, due dates, and completion status.

- **Performance Appraisals**:

- Formal review cycles with rating stars (1.0 to 5.0), KPI scores, self-assessments, and manager feedback.

- **HR Document Vault**:

- Secure storage and administrative verification of employee documents (Aadhaar, PAN, Offer Letters, Appointment Letters).

C. Admissions CRM & Telecalling Station

- **Omnichannel Ingestion**: Meta Lead Ads, Google Ads Webhook, Justdial Webhook, and website contact forms ingesting into `Lead` records.

- **Auto-Routing Engine**: Automatic round-robin or manual assignment to Counselors and Telecallers.

- **Telecaller Dashboard** (`/telecaller/dashboard`):

- Isolated workstation with daily call queues, instant call status updates (`INTERESTED`, `NOT\_INTERESTED`, `CALL\_BACK`, `FOLLOW\_UP`), and conversion tracking.

- **Counselor Desk** (`/counselor/dashboard`, `/counselor/leads`, `/counselor/admissions`):

- Lead qualification, scholarship/discount approval workflows, and 1-click conversion from Lead & Student Profile + Batch Enrollment.

D. Finance & Automated Fee Management

- **Tuition Fee Structures**: Flexible fee installment plans (`FULL\_PAYMENT`, `INSTALLMENTS`), custom due dates, and late fee policies.

- **Payment Processing**: Integrated Razorpay online checkout (`/api/webhooks/razorpay`) and offline transaction recording (Cash, Cheque, Bank Transfer).

- **Automated Ledger**: Student fee breakdown, balance calculations, automated receipt generation, and transaction audit trails.

E. Course CMS, LMS, Exams & Certification

- **Course CMS**: Multi-tier course curriculum hierarchy (Courses & Modules & Lessons).
- **Student LMS** (`/student/courses`, `/student/learning/[enrollmentId]`):
 - Interactive lesson viewer, video lectures, assignments, and reading materials with automatic progress tracking.
- **Examination & Assessment Engine** (`/student/exams`):
 - Question banks (MCQs, code snippets, theory), timed exam sessions, automated grading, percentage calculations, and passing flags.
- **Digital Certificates** (`/verify/certificate/[certificateNumber]`):
 - Automated certificate issuance upon course and exam completion with public verification URLs and QR validation.

5. DATABASE ARCHITECTURE (KEY PRISMA MODELS)

- **User & Auth**: `User`, `Account`, `Session`, `VerificationToken`, `AuditLog`.
- **Academics**: `Course`, `Module`, `Lesson`, `Batch`, `BatchTrainer`, `Enrollment`, `LessonProgress`, `ScheduledClass`, `AttendanceRecord`, `AttendanceEntry`.
- **CRM**: `Lead`, `FollowUpHistory`, `LeadActivity`, `AdmissionApplication`.
- **Finance**: `FeeStructure`, `FeeInstallment`, `PaymentTransaction`.
- **Assessments**: `Exam`, `QuestionBank`, `ExamAttempt`, `ExamAnswer`, `Certificate`.
- **Placement**:
 - `CorporatePartner`: Company profile and recruiter relations.
 - `RecruiterContact`: Individual company recruiters.
 - `RecruiterFollowUp`: Recruiter interaction history.
 - `JobDrive`: Campus drives and opening requirements.
 - `StudentPlacementProfile`: 1:1 career profile with resume, skills, and placement status.
 - `PlacementApplication`: Student application connected to a `JobDrive`.
 - `InterviewRound`: Rounds (`TECHNICAL`, `HR`, `MANAGERIAL`, `FINAL`), feedback, and results.
 - `PlacementOffer`: Financial compensation details and offer letter.

- `PlacementJoiningRecord`: Verified employment details and start dates.
- ****Staff & HRMS****:
 - `StaffProfile`: 1:1 institutional profile with compensation, manager, and bank details.
 - `StaffDepartmentEntity`: Institutional departments.
 - `StaffDesignation`: Hierarchical job roles.
 - `StaffAttendance`: Daily punch sheet (`PRESENT`, `ABSENT`, `LATE`, `EXCUSED`).
 - `StaffTask`: Assigned tasks and progress tracking.
 - `StaffPerformanceReview`: Performance reviews and ratings.
 - `StaffDocument`: Verified HR employee records.
 - `LeaveRequest`: Leave applications and approval workflow.
 - `PayrollRecord`: Monthly salary slip and payment records.

6. SERVICE & ROUTE STRUCTURE

```
SOFTLAB-GLOBAL/ └─ prisma/ └─ schema.prisma # Unified institutional schema (50+ models)
└─ src/ └─ app/ └─ (public)/ # Marketing website, courses, about, contact └─
admin/ # Admin hub (placements, staff, finance, courses, leads) └─ student/ # Student LMS,
learning player, exams, placements └─ trainer/ # Trainer portal, assigned batches, live classes └─
└─ counselor/ # Admissions desk, leads pipeline, follow-ups └─ telecaller/ # Isolated
telecalling desk └─ staff/ # Staff self-service (leave requests, salary slips) └─ api/ # tRPC
endpoint, NextAuth, and external webhooks └─ components/ └─ admin/placements/ # 7-
tab Placement Command Center └─ admin/staff/ # 7-tab Staff Management & HRMS Hub └─
└─ student/placements/ # Student Placement Portal & Profile Editor └─ common/sidebar.tsx
# Role-aware navigation sidebar └─ server/ └─ auth/rbac.ts # Role-Based Access Control
guards & aliases └─ services/ # Business logic services (Placement, Staff, CRM, Payroll, etc.) └─
└─ placement-enterprise.service.ts └─ staff-hrms.service.ts └─ job-drive.service.ts
└─ staff-profile.service.ts └─ payroll.service.ts └─ trpc/routers/ # Type-safe
tRPC routers └─ placement.ts # 25+ placement procedures └─ staff-erp.ts # 20+ HRMS
and payroll procedures └─ ...
```

7. CURRENT BUILD & VERIFICATION STATUS

1. **Prisma Schema**: Fully validated and pushed to Neon PostgreSQL (`npx prisma db push` succeeded).
2. **TypeScript Health**: `npx tsc --noEmit` exits with code `0` (Zero compiler errors).
3. **Relational Test Scripts**: Automated script verified talent pool extraction (live attendance % and exam scores), dynamic departments, and placement metrics against live database.
4. **Production Build**: `npm run build` completed successfully with code `0`, generating all 56 static and dynamic routes.
5. **Secret Hygiene**: Zero hardcoded credentials, API keys, or environment passwords in source code.

8. INSTRUCTIONS FOR ANY ASSISTANT (CHATGPT / CLAUDE / ANTIGRAVITY)

- **Do NOT** rebuild from scratch or create duplicate parallel models.
- **Do NOT** use mock or dummy state; always connect frontend components to existing tRPC routers in `src/server/trpc/routers/`.
- All monetary values in the database are stored as **integers** (either in Paise or Rupees as annotated in `schema.prisma`), formatted cleanly using Indian numbering format (₹ / LPA).
- Any new features must maintain non-destructive migrations on the existing Neon PostgreSQL database.

Upper jo images di gyi hai unke according or jo meri yeh summary hai inko combined kro or

Yeh Integrated nehi hai proper or razor pay ki jagah stripe ko inbuilt kro ya kisi ko bhi jo payment gateway start ho jaye baki iski proper summary na koi editable hai na main kuch kr sakta hu Yeh rhi aapke **SoftLab Global LMS + Marketing + Lead CRM** project ki shuruat se abhi tak ki **complete, detailed aur structured summary**, jise aap direct **ChatGPT** me paste kar sakte hain:

🚀 SOFTLAB GLOBAL LMS & MARKETING CRM — COMPLETE PROJECT MASTER SUMMARY

1. PROJECT OVERVIEW & TECH STACK

* **Project Name** : SoftLab Global LMS & Marketing CRM

* **Domain** : `https://www.softlabglobal.com/`

* **Core Stack** :

* **Framework** : Next.js 14.2.15 (App Router, Server Actions, Route Handlers)

* **Frontend** : React 18, Tailwind CSS, Lucide Icons, Shadcn UI

* **Backend API** : tRPC v11 (`@trpc/server` , `@trpc/client` , React Query v5)

* **Database & ORM** : PostgreSQL (Hosted on Neon AWS us-east-2) with Prisma ORM 5.22.0

* **Auth & RBAC** : NextAuth v5 (Beta 25) with 12 distinct system roles

* **Data Export** : `exceljs` (real `xlsx` generation) & RFC 4180 CSV engine

* **Key Operating Roles** :

1. `SUPER\_ADMIN` | 2. `DIRECTOR` | 3. `ADMIN` | 4. `MANAGER` | 5. `COUNSELOR` | 6. `TELECALLER` |
7. `MARKETING` | 8. `TRAINER` | 9. `HR` | 10. `ACCOUNTANT` | 11. `PLACEMENT\_OFFICER` | 12.
`STUDENT`

2. KEY INTEGRATIONS & BUSINESS CONFIGURATION

A. Meta / Facebook & Instagram Integration

* **Facebook Page ID** : `1322487780950878`

* **Target Boost Ad ID** : `1324192984113691`

* **Direct Meta Ad Center Link** :

`https://www.facebook.com/ad\_center/manage/?boost\_id=1324192984113691&entry\_point=www\_ad\_center\_overview\_ad\_cards&page\_id=1322487780950878`

* **Meta Graph API Version** : `v21.0`

* **Meta Conversions API (CAPI)** : Server-to-server offline conversion dispatch with SHA-256 hashed user phone/email.

* **Meta Inbound Webhook** : `https://www.softlabglobal.com/api/webhooks/meta` (Verify token: `softlab\_meta\_leads\_webhook\_2026\_verify`)

B. Other Ad Platforms & Lead Channels (14 Channels Total)

1. **Google Ads & Search**: Webhook key validation (`/api/webhooks/google-ads``), GCLID click tracking, offline conversion preparation.
2. **LinkedIn Ads**: Professional Lead Gen forms & sponsored campaign metrics.
3. **Justdial**: Live webhook ingestion (`/api/webhooks/justdial``).
4. **WhatsApp Cloud API & AI Bot**: 15-second auto-responder, qualification questionnaire (`/api/webhooks/whatsapp``).
5. **Website Channels**: Career Consultation Popup, Course Syllabus Inquiries, Contact Form, Walk-in, Student Referrals, Campus Drives, Organic Search & Social.

3. DATABASE SCHEMA ENHANCEMENTS (Prisma & Neon PostgreSQL)

Database schema (`prisma/schema.prisma``) me naye models aur enums add karke Neon PostgreSQL database me successfully push kar diye gaye hain (`npm run prisma db push --accept-data-loss`` executed, 0 errors):

1. **Enum Updates**:

- * `UserRoleCode``: `MARKETING`` role add kiya gaya.

- * `LeadSource``: `LINKEDIN``, `LINKEDIN_ADS`` add kiya gaya.

- * New Enums: `MarketingPlatform`` (`META``, `GOOGLE``, `LINKEDIN``, `JUSTDIAL``, `WHATSAPP``, `ORGANIC``, `OTHER``), `CampaignStatus`` (`ACTIVE``, `PAUSED``, `ARCHIVED``, `COMPLETED``, `DRAFT``), `AlertSeverity`` (`INFO``, `WARNING``, `CRITICAL``).

2. **New Normalized Marketing Models**:

- * `MarketingPlatformAccount``: Platform credentials, encrypted tokens, Ad Account IDs, Pixel IDs, sync health.

- * `MarketingCampaign``: Live/cached campaigns, budgets, spend, impressions, clicks, leads count, CPL, ROAS, revenue.

- * `MarketingAdSet``: Ad sets with targeting metadata and performance metrics.

- * `MarketingAd``: Ad creatives with preview URLs, spend, and lead attribution.

- * `MarketingLeadForm``: Lead Gen forms sync.

- * `LeadAttribution`: 1-to-1 relation with `Lead` storing `utmSource`, `utmMedium`, `utmCampaign`, `utmContent`, `utmTerm`, `fbclid`, `gclid`, `liFatId`, `landingPageUrl`, `referrerUrl`, first-touch vs last-touch timestamps.

- * `LeadScoreRule`: Dynamic lead scoring rules (weights, conditions, categories).

- * `OfflineConversionEvent`: Meta CAPI / Google Ads offline conversion events dispatch log (`Lead`, `Schedule`, `Purchase`/`Admitted`).

- * `MarketingAlert`: System alerts for High CPL, lead drop, webhook downtime, expired tokens.

- * `MarketingExportJob`: Audit log of all XLSX/CSV exports generated with download URLs and filters.

3. **Enhancements to Existing Models**:

- * `Lead`: Added `score Int @default(50)` (0-100 quality score) and `attribution LeadAttribution?`.

- * `User`: Added `resolvedMarketingAlerts` and `marketingExportJobs` relations.

4. BACKEND SERVICES IMPLEMENTED (`src/server/services/`)

1. **`lead-quality.service.ts`**:

- * Dynamic scoring engine (0 to 100) evaluating demographic data (clean 10-digit mobile), inbound ad channel intent (Meta Ads +15, Google Intent +20, LinkedIn +20, Referral +25), AI WhatsApp chatbot pre-screening (+20), and admissions stage progression.

- * Auto-assigns quality tiers:

- * 🔥 **HOT (75-100)**: Instant 5-min counselor call.

- * ⚡ **WARM (45-74)**: Telecaller qualification queue.

- * ❄️ **COLD (20-44)**: Nurturing sequence.

- * 🚫 **INVALID (<20)**: Excluded from active quota.

2. **`meta-marketing.service.ts`**:

- * Official Meta Graph API v21.0 client.

- * Fetches live campaigns with insights (`spend`, `impressions`, `clicks`, `leads`, `cpl`).

- * Updates campaign status (`ACTIVE`, `PAUSED`, `ARCHIVED`).

- * Creates real lead generation campaigns directly on Meta.

- * Sends real-time server-side offline conversions via Meta Conversions API (CAPI) with SHA-256 hashing.

- * Graceful fallback to local cached records when access tokens are pending.

3. **google-marketing.service.ts**:

- * Google Ads webhook key validation, GCLID click attribution, offline conversion dispatch prep.

4. **linkedin-marketing.service.ts**:

- * LinkedIn Lead Gen forms sync and Sponsored Content metrics tracking.

5. **marketing-export.service.ts**:

- * Enterprise data export engine producing genuine styled Excel (`.xlsx`) via `exceljs` and standard RFC 4180 `.csv`.

- * Exports all 24 required enterprise columns (Lead ID, Name, Phone, Email, City, Qualification, Course, CRM Status, Quality Rating, Score, Source, Platform, Campaign, AdSet, Creative, UTM Source, Medium, Campaign, Content, Term, Click ID, Telecaller, Counselor, Created At).

6. **marketing-alert.service.ts**:

- * Anomaly detection engine checking for high CPL surge (> ₹500), lead volume drop, and auth token expiration.

7. **marketing-sync.service.ts**:

- * Aggregates executive command center metrics: Total Leads, Total Ad Spend (₹), Blended CPL (₹), Admissions Attributed, Conversion %, Tuition Inflow Attributed (₹), ROAS, and the full 14-Channel Matrix breakdown.

5. CLIENT-SIDE UTM ATTRIBUTION & PUBLIC FORM TRACKING

1. **src/lib/utm.ts**:

- * Client-side utility that extracts `utm\_source`, `utm\_medium`, `utm\_campaign`, `utm\_content`, `utm\_term`, `fbclid`, `gclid`, `li\_fat\_id`, landing page URL, and document referrer from the visitor session.

- * Persists data in `sessionStorage` so navigation across pages does not lose original ad click attribution.

2. **Integrated Public Entry Points**:

- * `src/components/public/hero-lead-form.tsx` (Homepage banner form)

- * `src/components/public/career-counseling-modal.tsx` (Popup counseling modal)
- * `src/components/public/public-enquiry-form.tsx` (Course syllabus enquiry form)
- * `src/app/actions/enquiry.ts` (Next.js server action)
- * `src/server/services/crm-lead.service.ts` (`submitPublicEnquiry` auto-creates `LeadAttribution` and triggers initial quality score computation)
- * `src/server/trpc/routers/crm.ts` (`crm.submitEnquiry` mutation)

6. MARKETING tRPC ROUTER & RBAC NAVIGATION

1. **src/server/trpc/routers/marketing.ts**:

- * Registered in `src/server/trpc/routers/\_app.ts` as `api.marketing.\*`.
- * Accessible by: `SUPER\_ADMIN`, `DIRECTOR`, `ADMIN`, `MARKETING`, `MANAGER`.
- * Procedures:
 - * `getCommandCenterMetrics`: Executive KPIs & 14-channel breakdown.
 - * `getPlatformOverview`: Platform connection & health statuses.
 - * `getMetaCampaigns`: Live/cached Meta campaigns.
 - * `updateMetaCampaignStatus`: Toggle active/paused.
 - * `createMetaCampaign`: Create lead gen campaign.
 - * `getGoogleCampaigns` & `getLinkedInCampaigns`: Synced campaigns.
 - * `listAttributedLeads`: Paginated leads with multi-touch attribution & quality filter.
 - * `getLeadQualityRules` & `updateLeadQualityRule`: View & modify scoring weights.
 - * `recalculateLeadScore`: Instant rescore trigger.
 - * `exportLeadsData`: Returns binary base64 for `.xlsx` or raw string for `.csv`.
 - * `listAlerts` & `resolveAlert`: Notification and resolution of marketing alerts.
 - * `savePlatformCredentials`: In-app credentials save without modifying server `.env`.
 - * `triggerPlatformSync`: Manual one-click cross-platform refresh.

2. **Sidebar Navigation** (`src/components/common/sidebar.tsx`):

* Dedicated navigation items added for `MARKETING` role: Command Center, Meta Ads Manager, 14-Platform Matrix, Attribution Leads, Data Exports, Integrations & Setup.

7. MARKETING COMMAND CENTER UI (`src/components/admin/marketing/`)

Created a unified 7-tab command center inside `/admin/marketing` (`MarketingAnalyticsView`):

1. **Tab 1 — Command Center (`overview`)**:

- * High-level KPI cards (Total Leads, Admissions Closed, Conversion %, AI Bot Status).
- * Official Instagram (@softlabglobal9) live feed & engagement metrics.
- * Counselor & Telecaller communication lines and lead capacity routing.
- * Real-time automated marketing activities feed.

2. **Tab 2 — Meta Ads Manager (`meta`)**:

- * Facebook Boost Ad card (`#1324192984113691`) with direct link to Meta Ad Center.
- * Real-time campaign table with budget, spend, impressions, clicks, leads, CPL, ROAS.
- * Play/Pause campaign status toggle button.
- * "Launch Campaign" modal.
- * Meta Conversions API (CAPI) server-to-server health status.

3. **Tab 3 — 14-Platform Matrix (`platforms`)**:

* Complete channel matrix showing Lead Count, Lead Share %, Ad Spend, CPL, Admissions, Tuition Revenue, and Status (`CONNECTED` / `ORGANIC` / `PENDING\_CONFIG`).

4. **Tab 4 — Attributed Leads (`leads`)**:

* Searchable and filterable table displaying leads with quality score badges (🔥 HOT, ⚡ WARM, ❄️ COLD, INVALID), UTM parameters, click IDs (`fbclid`, `gclid`), assigned telecaller/counselor, and instant rescore trigger.

5. **Tab 5 — Lead Quality Rules (`scoring`)**:

* Interactive rule weights configuration table (Valid Mobile +20, Meta Ad +15, Google Search +20, LinkedIn +20, Referral +25, AI Bot Qualified +20, etc.) with active/disabled toggles.

6. **Tab 6 — Data Exports (`export`)**:

- * Modal to generate styled `.xlsx` or `.csv` files with custom date range, source, quality rating, and status filters. Triggers direct browser download.

7. **Tab 7 — Credentials & Webhooks (`integrations`)**:

- * Single consolidated checklist for external platform setup.
- * Webhook URLs with 1-click copy for Meta, Google Ads, Justdial, and WhatsApp.
- * In-app credential save form for Meta Ad Account ID, Pixel ID, and Access Token.

8. CONSOLIDATED EXTERNAL CREDENTIAL CHECKLIST

Jab bhi client/admin ko external platforms se live sync verify karna ho, ye single checklist use hogi:

| Platform | Required Keys | Production Endpoint / Callback URL |

| :--- | :--- | :--- |

| **Meta / Facebook Ads** | `META\_APP\_ID`
(System User Long-Lived Token)
(`1322487780950878`)
`META\_APP\_SECRET`
`META\_ACCESS\_TOKEN`
`META\_AD\_ACCOUNT\_ID`
(`act\_...`)
`META\_PAGE\_ID`
`META\_PIXEL\_ID`
`META\_CONVERSION\_TOKEN` (CAPI) | **Webhook**
Callback URL
`https://www.softlabglobal.com/api/webhooks/meta`
Verify
Token
`softlab\_meta\_leads\_webhook\_2026\_verify` |

| **Google Ads** |
`GOOGLE\_ADS\_CUSTOMER\_ID`
`GOOGLE\_ADS\_DEVELOPER\_TOKEN`
`GOOGLE\_ADS\_CLIENT\_ID`
`GOOGLE\_ADS\_CLIENT\_SECRET`
`GOOGLE\_ADS\_REFRESH\_TOKEN` | **Webhook** Callback
URL
`https://www.softlabglobal.com/api/webhooks/google-ads`
Key
Header
`softlab-google-ads-key-2026` |

| **LinkedIn Ads** |
`LINKEDIN\_AD\_ACCOUNT\_ID`
`LINKEDIN\_ACCESS\_TOKEN`
`LINKEDIN\_ORGANIZATION\_ID` |
OAuth 2.0 Lead Gen Sync |

| **Justdial** | Inbound Lead API / Webhook | **Webhook**
URL
`https://www.softlabglobal.com/api/webhooks/justdial` |

| **WhatsApp Cloud API** |
`WHATSAPP\_PHONE\_NUMBER\_ID`
`WHATSAPP\_ACCESS\_TOKEN`
`WHATSAPP\_VERIFY\_TOKEN` | **Webhook** URL
`https://www.softlabglobal.com/api/webhooks/whatsapp` |

9. CURRENT STATUS & WHAT'S COMPLETED

- ☒ **Database**: 100% in sync with Neon PostgreSQL (``npx prisma db push --accept-data-loss`` succeeded, all 10 marketing tables created).
- ☒ **Dependencies**: ``exceljs`` installed and operational.
- ☒ **Backend**: All 6 marketing services and ``marketingRouter`` created and mounted in ``trpc`` ``appRouter``.
- ☒ **Attribution Tracking**: ``src/lib/utm.ts`` and all public forms/actions updated with automatic UTM and click ID capture.
- ☒ **Command Center UI**: 7 tabs fully built and integrated into ``/admin/marketing``.
- ☒ **Role Authorization**: ``MARKETING`` role added to Prisma schema, sidebar, and admin marketing page.
- ☒ **Build & Integrity**: All existing counselor, telecaller, student, and LMS features remain completely intact without any breaking changes.

Is module ko proper run krana hai

You are now the LEAD SOFTWARE ARCHITECT, SENIOR FULL-STACK ENGINEER, UI/UX DESIGNER, DATABASE ARCHITECT, SECURITY ENGINEER, QA ENGINEER and DEVOPS ENGINEER for my existing LMS.

Your job is to transform my existing LMS into a complete, production-ready, modern LMS + HRMS + Staff Management platform.

IMPORTANT:

Do not treat this as a new standalone application.

You MUST inspect, understand and integrate with the existing LMS architecture, database, authentication, users, roles, courses, batches, trainers, students, attendance, dashboards and existing modules.

Do not unnecessarily rebuild or duplicate existing functionality.

PHASE 0 — FULL PROJECT DISCOVERY

Before making major changes, inspect the entire project.

Analyze:

- Frontend framework
- Backend framework
- Database
- ORM
- Authentication
- Authorization
- Existing roles
- Existing users
- Existing admin panel
- Existing LMS dashboard
- Students
- Trainers
- Courses
- Batches
- Attendance
- Fees/payment modules
- Reports
- Notifications
- File storage
- APIs
- Routing
- Components
- State management
- Database relationships
- Existing UI design system
- Environment configuration
- Deployment configuration

Find reusable components and services.

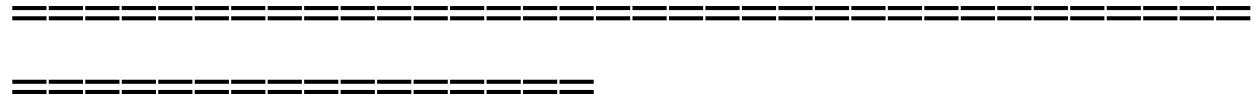
Identify:

- What already exists
- What can be reused
- What must be extended

- What is missing
- Potential conflicts
- Duplicate functionality
- Database risks
- Security risks
- Performance risks

Create an internal implementation plan based on the actual codebase.

DO NOT destroy or replace working LMS functionality.



PHASE 1 — ARCHITECTURE

Design an integrated HRMS architecture inside the existing LMS.

The HRMS must be connected with existing:

- Users
- Trainers
- Employees
- Students
- Courses
- Batches
- Attendance
- Departments
- Roles
- Notifications
- Reports

Create normalized and scalable database structures.

Where an existing table/model can be reused safely, reuse it.

Avoid duplicate employee/user records.

The Employee entity should be properly connected to the existing authentication/user system.

Create appropriate relationships for:

Employee
Department

Designation
Branch
Role
Attendance
Leave
Shift
Salary
Payroll
Performance
KPI
Tasks
Training
Assets
Documents
Recruitment
Onboarding
Offboarding
Notifications
Audit Logs

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PHASE 2 — HRMS MAIN DASHBOARD

Create a premium, modern enterprise HRMS dashboard.

It must look like a professional SaaS/ERP product.

Dashboard should contain:

TOP KPI CARDS:

- Total Employees
- Active Employees
- Inactive Employees
- New Employees
- Present Today
- Absent Today
- On Leave Today
- Late Today
- Pending Leave Requests
- Pending Tasks

Each KPI card must:

- Show real database data
- Be clickable
- Navigate to the related module
- Support permissions
- Show meaningful trends where data exists

Create interactive visualizations:

1. Employee Growth — Line Chart
2. Attendance Overview — Donut/Pie Chart
3. Department Distribution — Bar Chart
4. Leave Trends — Line Chart
5. Employee Status — Chart
6. Joining vs Exit — Line Chart
7. Staff Performance — Bar Chart
8. Payroll Overview — Line Chart
9. Task Completion — Chart
10. Training Progress — Chart

Create widgets:

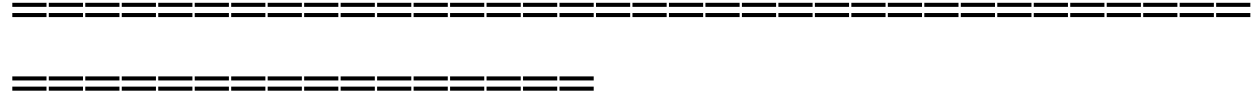
- Today's Birthdays
- Upcoming Birthdays
- Work Anniversaries
- Employees on Leave
- Pending Approvals
- Recent HR Activities
- Upcoming Interviews
- Expiring Documents
- Probation Ending Soon
- Contract Expiry
- Overdue Tasks

Dashboard filters:

- Today
- This Week
- This Month
- This Quarter
- This Year
- Custom Date Range
- Department
- Branch
- Employee Type

Charts MUST use actual database data.

Do not hard-code fake chart values once real data exists.



PHASE 3 — EMPLOYEE / STAFF MANAGEMENT

Create complete Employee Management.

Features:

- Employee Directory
- Add Employee
- Edit Employee
- View Employee
- Deactivate Employee
- Reactivate Employee
- Archive Employee
- Employee Search
- Advanced Filters
- Bulk Actions
- Import CSV/Excel
- Export CSV/Excel/PDF

Employee profile must include:

PERSONAL:

- Employee ID
- Profile Photo
- Full Name
- Gender
- Date of Birth
- Mobile
- Alternate Mobile
- Email
- Address
- City
- State
- PIN

- Emergency Contact
- Emergency Contact Number

EMPLOYMENT:

- Joining Date
- Department
- Designation
- Employee Type
- Employee Status
- Branch
- Reporting Manager
- Shift
- Work Mode
- Grade
- Probation Period
- Confirmation Date
- Notice Period
- Last Working Date

PROFESSIONAL:

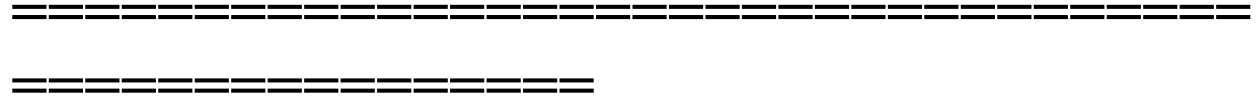
- Qualification
- Specialization
- Experience
- Previous Company
- Previous Designation
- Skills
- Technical Skills
- Soft Skills
- Certifications
- Languages
- Resume

DOCUMENTS:

- Identity Document
- PAN
- Address Proof
- Resume
- Educational Certificates
- Experience Certificates
- Offer Letter
- Appointment Letter
- Joining Documents
- Bank Documents

- Other Documents

All appropriate fields must be editable.



PHASE 4 — EMPLOYEE PROFILE

Create a premium employee profile page.

Header:

Profile Photo
Employee Name
Employee ID
Designation
Department
Status

Tabs:

- Overview
- Personal
- Employment
- Attendance
- Leave
- Payroll
- Performance
- Tasks
- Training
- Assets
- Documents
- Notes
- Activity

Employee statistics:

- Attendance %
- Leave Balance
- Tasks Completed
- Performance Score
- Training Completed
- Experience

- Salary
- Pending Tasks

PHASE 5 — DEPARTMENT & DESIGNATION

Create fully editable:

Department Management
Designation Management
Branch Management
Employee Category Management
Employee Grade Management

Admin must be able to:

- Add
- Edit
- Delete where safe
- Activate
- Deactivate
- Assign employees
- Assign managers

Nothing should be unnecessarily hard-coded.

PHASE 6 — ATTENDANCE

Create complete Attendance Management.

Support:

- Check-In
- Check-Out
- Present

- Absent
- Half Day
- Leave
- Holiday
- Weekly Off
- Late
- Early Exit
- Overtime
- Working Hours

Views:

- Today's Attendance
- Daily Attendance
- Monthly Attendance
- Employee Attendance
- Department Attendance

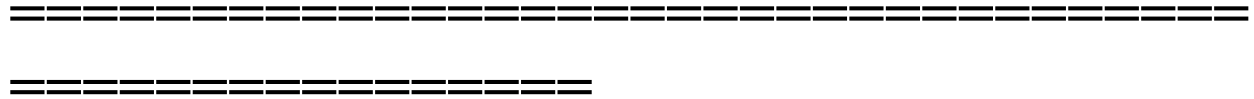
Filters:

- Date
- Employee
- Department
- Branch
- Status
- Shift

Create attendance analytics.

Allow authorized administrators to correct attendance.

Every manual correction must be audit logged.



PHASE 7 — LEAVE MANAGEMENT

Create complete Leave Management.

Admin-configurable leave types:

- Casual
- Sick

- Earned
- Paid
- Unpaid
- Emergency
- Half Day
- Custom Leave Types

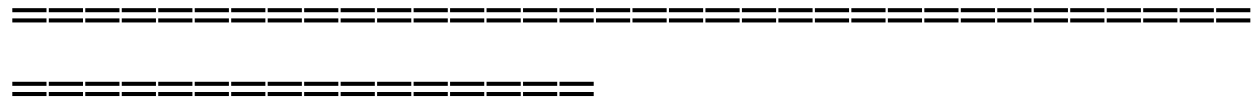
Features:

- Apply Leave
- Approve
- Reject
- Cancel
- Leave Balance
- Leave History
- Leave Calendar
- Pending Requests

Approval workflow:

Employee → Reporting Manager → HR/Admin

Make workflow configurable.



PHASE 8 — SHIFT MANAGEMENT

Create configurable shifts.

Fields:

- Shift Name
- Start Time
- End Time
- Break Time
- Grace Period
- Working Hours
- Weekly Off
- Status

Employees can be assigned different shifts.

PHASE 9 — PAYROLL

Create a complete Payroll Management system.

EARNINGS:

- Basic
- HRA
- Conveyance
- Special Allowance
- Incentive
- Bonus
- Overtime
- Other Earnings

DEDUCTIONS:

- PF
- ESI
- Professional Tax
- TDS
- Loan
- Advance
- Other Deductions

Calculate:

Gross Salary

Total Deductions

Net Salary

Support:

- Salary Structures
- Employee Salary History
- Monthly Payroll
- Payroll Processing
- Payslips
- Payroll Reports
- Department Payroll
- Employee Payroll

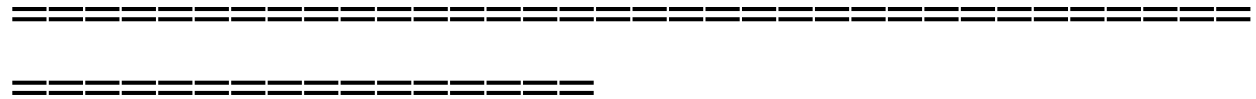
Generate professional payslips.

IMPORTANT:

Do not hard-code country-specific statutory calculations unless the existing application already has verified rules/configuration.

Make statutory components configurable.

Restrict payroll information using server-side permissions.



PHASE 10 — PERFORMANCE MANAGEMENT

Create:

- Performance Reviews
- KPI Management
- Goals
- Employee Evaluation
- Manager Evaluation
- Self Evaluation
- Performance History

Metrics:

- Attendance
- Productivity
- Quality
- Task Completion
- Communication
- Teamwork
- Technical Skills
- Customer Handling
- Discipline
- Overall Performance

Support:

Monthly
Quarterly
Half-Yearly
Yearly

Admin must be able to create custom KPIs.

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PHASE 11 — STAFF TASK MANAGEMENT

Create Task Management.

Fields:

- Task
- Description
- Employee
- Department
- Priority
- Due Date
- Status
- Progress
- Attachment
- Comments

Statuses:

- Pending
- In Progress
- Completed
- On Hold
- Cancelled

Dashboard:

- Total Tasks
- Completed
- Pending
- Overdue

PHASE 12 — TRAINING & DEVELOPMENT

Create:

- Training Programs
- Training Calendar
- Employee Training
- Trainers
- Skills
- Certificates
- Assessments
- Completion Status

Analytics:

- Training Completed
 - Training Pending
 - Training Hours
 - Assessment Scores
 - Skill Development
-
-

PHASE 13 — ASSET MANAGEMENT

Create Staff Asset Management.

Support:

- Laptop
- Desktop
- Monitor
- Keyboard
- Mouse
- Mobile
- SIM

- ID Card
- Access Card
- Headset
- Other Assets

Fields:

Asset ID
 Asset Name
 Category
 Serial Number
 Employee
 Issue Date
 Return Date
 Condition
 Status
 Remarks

Maintain complete assignment history.

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PHASE 14 — RECRUITMENT

Create Recruitment module.

Pipeline:

Applicant
 → Screening
 → Interview
 → Selected
 → Offer
 → Joined
 → Active Employee

Features:

- Candidate Profile
- Resume
- Interview
- Interview Feedback
- Offer

- Joining
- Document Verification
- Onboarding

PHASE 15 — ONBOARDING

Create configurable onboarding checklist.

Examples:

- Documents
- ID Creation
- Email Account
- System Access
- Assets
- Orientation
- Department Introduction
- Training

Show onboarding completion percentage.

PHASE 16 — OFFBOARDING

Create:

- Resignation
- Notice Period
- Last Working Date
- Exit Interview
- Asset Return
- Document Clearance
- Salary Clearance
- Final Settlement
- Exit Reason

Maintain complete history.

PHASE 17 — DOCUMENT & LETTER GENERATOR

Create editable document templates.

Templates:

- Offer Letter
- Appointment Letter
- Joining Letter
- Experience Letter
- Relieving Letter
- Salary Certificate
- Warning Letter
- Promotion Letter
- Increment Letter
- Internship Certificate
- Custom Letters

Support variables:

{{EmployeeName}}
{{EmployeeID}}
{{Designation}}
{{Department}}
{{JoiningDate}}
{{Salary}}
{{ManagerName}}
{{CompanyName}}

Allow:

- Edit Template
- Preview
- Generate
- PDF
- Print

PHASE 18 — NOTIFICATION SYSTEM

Create notifications for:

- Leave Requests
- Leave Approval
- Attendance Issues
- Late Arrival
- Birthday
- Anniversary
- Document Expiry
- Probation Completion
- Contract Expiry
- Task Due
- Training
- Payroll
- HR Announcements

Create notification center with:

- Read
- Unread
- Mark all as read
- Notification preferences

PHASE 19 — HR CALENDAR

Create interactive calendar.

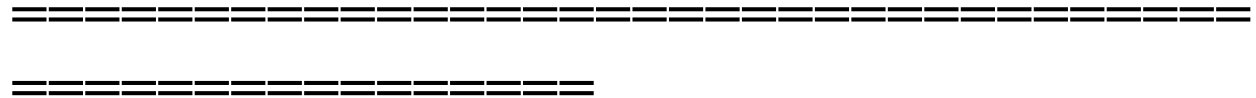
Events:

- Holidays
- Birthdays
- Anniversaries
- Leaves
- Training

- Interviews
- Meetings
- HR Events

Support:

Day
Week
Month



PHASE 20 — REPORTING & ANALYTICS

Create professional reporting system.

Reports:

- Employee
- Attendance
- Leave
- Payroll
- Performance
- Recruitment
- Attrition
- Training
- Assets
- Documents
- Tasks
- Salary
- Department

Reports must support:

- Search
- Filters
- Sorting
- Date range
- Column selection
- Export CSV
- Export Excel
- Export PDF
- Print

Create custom report capability if compatible with the existing architecture.

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PHASE 21 — ROLE BASED ACCESS CONTROL

Create granular RBAC.

Default roles:

- Super Admin
- HR Admin
- HR Manager
- Department Manager
- Team Leader
- Trainer
- Employee
- Accountant
- Support Staff

Permissions:

- View
- Add
- Edit
- Delete
- Approve
- Export
- Print

Permissions must be enforced on the SERVER/API level.

Do NOT rely only on hiding frontend buttons.

Allow administrators to create custom roles.

PHASE 22 — AUDIT LOG

Create complete audit logs.

Track:

- User
- Action
- Module
- Record
- Date
- Time
- Old Value
- New Value

Track sensitive operations such as:

- Salary changes
 - Permission changes
 - Employee changes
 - Attendance corrections
 - Leave approval
 - Payroll changes
 - Document changes
 - Role changes
-
-

PHASE 23 — GLOBAL SEARCH

Create global HRMS search.

Search:

- Employee
- Employee ID
- Mobile
- Email

- Department
- Designation
- Skills
- Tasks
- Documents

Implement appropriate indexes.

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PHASE 24 — IMPORT / EXPORT

Support:

Employee Import
Attendance Import
Department Import
Bulk Employee Update

Before importing:

1. Validate file
2. Preview records
3. Show errors
4. Show duplicate records
5. Allow confirmation
6. Import valid records
7. Generate import summary

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PHASE 25 — UI/UX

The HRMS must look PREMIUM.

Use:

- Modern SaaS design
- Responsive layout
- Professional sidebar

- Top navigation
- Breadcrumbs
- Cards
- Charts
- Tables
- Filters
- Search
- Tabs
- Modals
- Drawers
- Status badges
- Tooltips
- Skeleton loaders
- Empty states
- Error states
- Confirmation dialogs

Support:

Light Mode

Dark Mode

Maintain consistency with the existing LMS design.

Do not introduce an unrelated visual language.

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PHASE 26 — FULL EDITABILITY

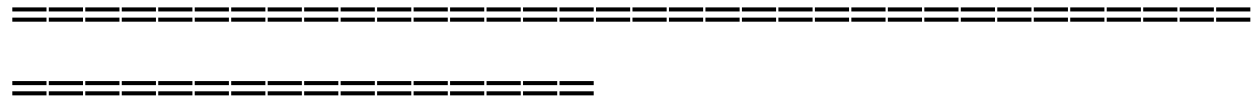
Avoid unnecessary hard-coded values.

Admin must be able to configure:

- Departments
- Designations
- Branches
- Employee categories
- Leave types
- Shifts
- Salary components
- Payroll settings
- Performance KPIs

- Task statuses
- Training categories
- Asset categories
- Document templates
- Notification templates
- Roles
- Permissions
- Organization settings

Support custom fields where technically appropriate.



PHASE 27 — SECURITY

Perform production-grade security implementation.

Protect:

- Authentication
- Authorization
- Employee data
- Salary
- Bank information
- Documents
- APIs
- File uploads
- Sessions
- Admin functions

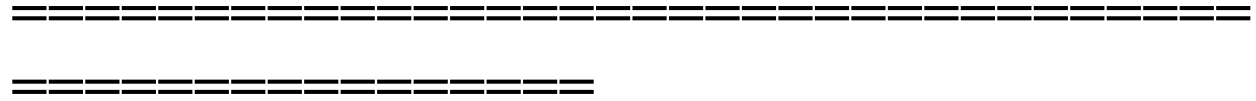
Check for:

- IDOR
- Privilege escalation
- SQL injection
- XSS
- CSRF
- Broken authorization
- Sensitive data exposure
- Unsafe file uploads
- Missing API authorization
- Frontend-only permission checks
- Rate limiting requirements

Never expose secrets.

Never hard-code credentials.

Never bypass existing authentication.



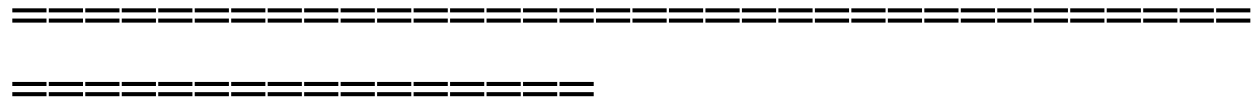
PHASE 28 — PERFORMANCE

Optimize the system.

Use appropriate:

- Database indexes
- Pagination
- Lazy loading
- Query optimization
- Caching where appropriate
- Debounced search
- Efficient API calls
- Component reuse

Do not load thousands of records unnecessarily.



PHASE 29 — TESTING

After implementation, test the entire system.

Run:

- Type checks
- Lint
- Unit tests where available
- Integration tests
- API tests
- Permission tests
- Database tests

- UI tests
- Browser tests

Use browser automation to test real user workflows.

Test:

HR Admin
Manager
Employee
Accountant
Trainer
Super Admin

Verify that each role sees only permitted data.

Test:

Employee creation
Employee editing
Attendance
Leave
Payroll
Performance
Tasks
Training
Assets
Documents
Recruitment
Onboarding
Offboarding
Reports
Notifications

Fix errors found during testing.

Then retest.

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PHASE 30 — BROWSER QA

Open the running LMS in the browser.

Perform complete end-to-end testing.

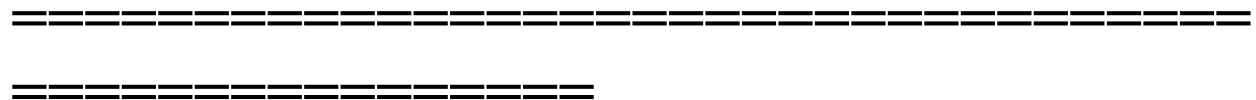
Check:

- Navigation
- Buttons
- Forms
- Search
- Filters
- Tables
- Pagination
- Modals
- Charts
- Mobile responsiveness
- Dark mode
- Permissions
- Error handling
- Loading states
- Empty states

Test real workflows instead of only checking whether pages load.

If an issue is found:

1. Identify root cause.
2. Fix it.
3. Retest.
4. Verify no regression.



PHASE 31 — DATA INTEGRITY

Verify that:

Employee
→ Department
→ Attendance
→ Leave
→ Payroll
→ Performance
→ Tasks
→ Training

- Assets
- Documents

remain correctly interconnected.

Avoid duplicated employee records.

Ensure deleting/deactivating an employee does not accidentally destroy required historical records.

Use soft-delete/archive where appropriate.

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PHASE 32 — DEMO DATA

Create realistic demo/sample data ONLY if the database is empty or demo mode is required.

Include:

- Employees
- Departments
- Designations
- Attendance
- Leave
- Tasks
- Training
- Assets
- Performance
- Payroll

Clearly distinguish demo data from real data.

Never overwrite existing production data.

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PHASE 33 — DEPLOYMENT SAFETY

Before applying database migrations or destructive changes:

- Check current schema
- Create safe migration
- Avoid destructive migration unless explicitly required
- Preserve existing data
- Provide rollback strategy

Do not change production environment variables unnecessarily.

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PHASE 34 — FINAL SYSTEM AUDIT

After completing implementation, perform a complete final audit.

Check:

FUNCTIONALITY

- All modules work

DATABASE

- Relationships correct
- No unnecessary duplication
- Indexes appropriate

SECURITY

- Permissions enforced server-side

UI

- Professional
- Responsive
- Consistent

PERFORMANCE

- No obvious inefficient queries

TESTING

- Major workflows tested

INTEGRATION

- HRMS works with existing LMS

REPORTING

- Real data shown

AUDIT

- Sensitive changes tracked

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CRITICAL DEVELOPMENT RULES

1. DO NOT destroy existing LMS functionality.
2. DO NOT replace the existing framework unless absolutely necessary.
3. DO NOT create duplicate systems where existing functionality can be extended.
4. DO NOT hard-code business data unnecessarily.
5. DO NOT use fake dashboard numbers when real database data exists.
6. DO NOT bypass authentication.
7. DO NOT bypass authorization.
8. DO NOT expose payroll or sensitive employee information.
9. DO NOT create insecure APIs.
10. DO NOT make destructive database changes without protection.
11. DO NOT modify unrelated modules unnecessarily.
12. Reuse existing components and architecture whenever practical.
13. Keep code modular and maintainable.
14. Use clear naming conventions.
15. Add appropriate comments only where they improve maintainability.
16. Make the system scalable for future expansion.
17. Every major feature must be testable.
18. Every major workflow must handle errors gracefully.
19. Every important admin action must be auditable.
20. If you encounter an architectural decision that could significantly affect existing LMS functionality, inspect the codebase and choose the least disruptive compatible approach.

FINAL OBJECTIVE

The final result should be a single integrated:

LMS + HRMS + STAFF MANAGEMENT + ATTENDANCE + LEAVE + PAYROLL +
PERFORMANCE + TASK + TRAINING + ASSET + RECRUITMENT + DOCUMENT +
REPORTING PLATFORM.

It must feel like a professional enterprise SaaS application.

Do not merely generate pages.

Build the actual functional system with:

Frontend
Backend
Database
APIs
Authentication
Authorization
Validation
Reports
Charts
Notifications
Audit Logs
Testing
Browser QA
Responsive UI
Dark Mode
Light Mode

At the end, provide a concise implementation report containing:

1. Modules completed
2. Files created
3. Files modified
4. Database tables/models created
5. APIs created
6. Permissions created
7. Tests performed
8. Browser tests performed

9. Issues discovered
10. Issues fixed
11. Remaining limitations
12. Deployment/migration instructions

If a feature cannot be safely implemented because the existing LMS architecture does not support it, do not fake the functionality. Explain the technical limitation and implement the safest compatible alternative.

Yeh student Management ke liye hai

You are the LEAD SOFTWARE ARCHITECT, SENIOR FULL-STACK ENGINEER, DATABASE ARCHITECT, UI/UX DESIGNER, SECURITY ENGINEER and QA ENGINEER for my existing LMS.

Build and integrate a complete, production-ready **STUDENT MANAGEMENT SYSTEM (SMS)** into my existing LMS.

IMPORTANT:

This is NOT a standalone application.

You MUST inspect and understand the existing LMS before implementation.

Integrate Student Management with existing:

- Users
- Authentication
- Roles
- Courses
- Batches
- Trainers
- Attendance
- Fees/Payments
- Exams
- Certificates
- Notifications
- Reports
- Existing dashboard
- Existing database
- Existing UI components

Do not duplicate existing functionality unnecessarily.

Do not break any existing LMS feature.

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1. PROJECT DISCOVERY

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First inspect the complete existing LMS.

Analyze:

- Frontend framework
- Backend framework
- Database
- ORM
- Authentication
- Authorization
- Existing student model
- Existing user model
- Course model
- Batch model
- Trainer model
- Attendance
- Fees
- Payments
- Exams
- Certificates
- Notifications
- Reports
- File storage
- APIs
- Routing
- Existing dashboards
- Existing UI design system

Determine:

- What already exists
- What can be reused
- What needs extension
- What is missing
- Existing database relationships
- Potential conflicts
- Security risks
- Data duplication risks

Do not modify major architecture until the existing system is understood.

2. STUDENT MANAGEMENT DASHBOARD

Create a premium, modern and visually attractive Student Management Dashboard.

Top KPI Cards:

- Total Students
- Active Students
- New Admissions
- Students in Training
- Completed Students
- Inactive Students
- Students on Hold
- Dropped Students
- Today's Present Students
- Today's Absent Students
- Pending Fees
- Upcoming Exams

Every KPI should:

- Use real database data
- Be clickable
- Navigate to relevant records
- Respect user permissions
- Support appropriate filters

Create interactive charts:

1. Student Growth — Monthly Line Chart
2. Admissions Trend — Monthly Line Chart
3. Course-wise Student Distribution — Bar Chart
4. Batch-wise Student Distribution — Bar Chart
5. Attendance Overview — Donut/Pie Chart
6. Fee Collection — Monthly Line/Bar Chart
7. Pending Fees — Chart
8. Student Status Distribution — Chart
9. Course Completion — Chart
10. Exam Performance — Chart

- 11. Dropout/Withdrawal Trend — Chart
- 12. Lead-to-Admission Conversion — Chart if CRM/Lead data exists

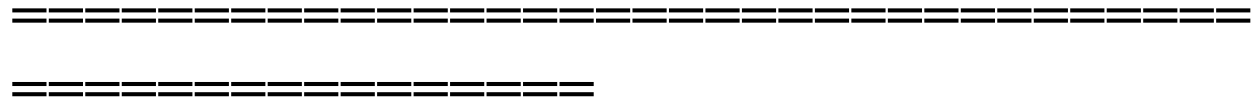
Dashboard widgets:

- Today's Admissions
- Recent Students
- Upcoming Birthdays
- Students with Pending Fees
- Students with Low Attendance
- Upcoming Exams
- Recent Payments
- Recently Completed Courses
- Recently Issued Certificates
- Students Requiring Attention
- Recent Activities

Dashboard filters:

- Today
- This Week
- This Month
- This Quarter
- This Year
- Custom Date Range
- Course
- Batch
- Trainer
- Branch
- Student Status

All charts must use actual database data.



3. STUDENT UNIQUE ID

Every student must have a unique Student ID.

Example:

STU-2026-00001

Student ID generation must be configurable.

Support:

- Automatic ID
- Manual ID where authorized
- Prefix configuration
- Academic/session year
- Sequential numbering

Prevent duplicate Student IDs.

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4. COMPLETE STUDENT PROFILE

Create a complete Student 360° profile.

Student profile header:

- Profile Photo
- Student Name
- Student ID
- Admission Number
- Status
- Course
- Batch
- Branch
- Admission Date
- Contact Number
- Email

Profile tabs:

1. Overview
2. Personal Details
3. Contact Details
4. Guardian Details
5. Academic Details
6. Admission Details
7. Course Details
8. Batch Details
9. Attendance

- 10. Fees
- 11. Payments
- 12. Exams
- 13. Results
- 14. Assignments
- 15. Certificates
- 16. Documents
- 17. Communication
- 18. Notes
- 19. Activity History

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5. PERSONAL DETAILS

Student fields:

- Student ID
- Admission Number
- First Name
- Middle Name
- Last Name
- Full Name
- Profile Photo
- Gender
- Date of Birth
- Age
- Blood Group
- Nationality
- Mother Tongue
- Category where applicable
- Identification details where legally required
- Marital Status where relevant
- Student Type

Do not make sensitive fields mandatory unless required by the existing LMS/business process.

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6. CONTACT DETAILS

Fields:

- Primary Mobile
- Alternate Mobile
- WhatsApp Number
- Email
- Alternate Email
- Address
- City
- State
- Country
- PIN Code

Support multiple addresses where appropriate:

- Permanent Address
- Current Address

Allow:

"Same as Permanent Address"

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7. PARENT / GUARDIAN DETAILS

Create complete guardian management.

Fields:

- Guardian Name
- Relationship
- Father Name
- Mother Name
- Guardian Mobile
- Alternate Mobile
- Email
- Occupation
- Company
- Annual Income where appropriate
- Address
- Emergency Contact
- Emergency Relationship

Allow multiple guardians.

Allow primary guardian selection.

8. ACADEMIC DETAILS

Fields:

- Highest Qualification
- School/College
- Board/University
- Passing Year
- Percentage/CGPA
- Stream
- Graduation
- Post Graduation
- Technical Qualification
- Certifications
- Skills
- Previous Training
- Work Experience
- Current Employment Status

Allow multiple academic records.

Example:

10th
12th
Diploma
Graduation
Post Graduation
Certification

9. ADMISSION MANAGEMENT

Create complete Admission Management.

Admission fields:

- Admission Number
- Student ID
- Admission Date
- Admission Type
- Admission Source
- Counsellor
- Branch
- Course
- Batch
- Trainer
- Session
- Academic Year
- Joining Date
- Expected Completion Date
- Admission Status
- Referral Source
- Remarks

Admission statuses:

- Enquiry
- Applied
- Counselling
- Selected
- Admission Pending
- Admitted
- Active
- Completed
- On Hold
- Withdrawn
- Dropped
- Transferred

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10. COURSE ENROLLMENT

A student can enroll in one or multiple courses.

Create Course Enrollment Management.

Fields:

- Student
- Course
- Enrollment ID
- Enrollment Date
- Start Date
- Expected End Date
- Completion Date
- Course Status
- Trainer
- Batch
- Branch
- Fee Plan
- Discount
- Final Fee
- Progress %
- Completion Status

Statuses:

- Enrolled
- Active
- On Hold
- Completed
- Withdrawn
- Dropped
- Transferred

Do not duplicate the student record for multiple courses.

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11. BATCH MANAGEMENT INTEGRATION

Connect students with existing Batch Management.

Show:

- Batch Name
- Batch ID
- Course
- Trainer
- Room
- Schedule
- Start Date
- End Date
- Student Count

Allow authorized users to:

- Add student to batch
- Transfer student
- Remove student
- Change batch
- View batch history

Maintain transfer history.

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12. STUDENT ATTENDANCE

Create complete Student Attendance.

Support:

- Present
- Absent
- Late
- Half Day
- Leave
- Holiday
- Excused

Attendance can be associated with:

- Course
- Batch
- Session
- Trainer
- Date

Show:

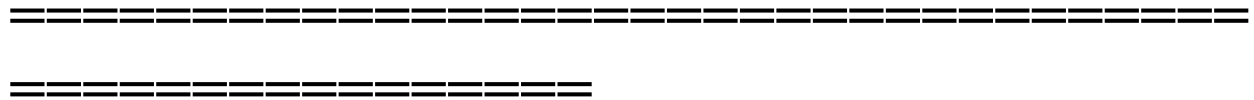
- Daily Attendance
- Monthly Attendance
- Course Attendance
- Batch Attendance
- Student Attendance
- Attendance Percentage

Create alerts for low attendance.

Make threshold configurable.

Example:

Attendance below configured percentage → Warning.



13. STUDENT FEES & FINANCE

Create complete Student Fee Management.

Fee structure:

- Course Fee
- Registration Fee
- Admission Fee
- Tuition Fee
- Exam Fee
- Certification Fee
- Material Fee
- Lab Fee
- Other Fee

Support:

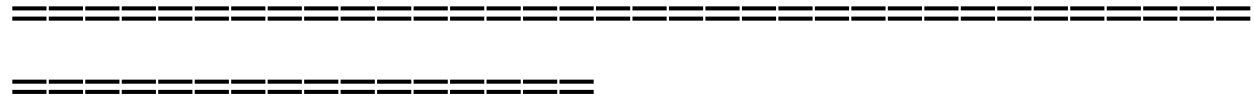
- Discount
- Scholarship
- Coupon
- Adjustment
- Tax where applicable
- Late Fee

Automatically calculate:

Total Fee
Discount
Paid Amount
Pending Amount
Overdue Amount

Create:

- Fee Schedule
- Installments
- Payment History
- Pending Fees
- Overdue Fees
- Refunds
- Adjustments



14. PAYMENT MANAGEMENT

Payment fields:

- Payment ID
- Student
- Invoice
- Amount
- Payment Date
- Payment Method
- Transaction ID
- Reference Number
- Collected By
- Remarks
- Status

Payment methods:

- Cash
- Bank Transfer
- UPI
- Card
- Online Payment

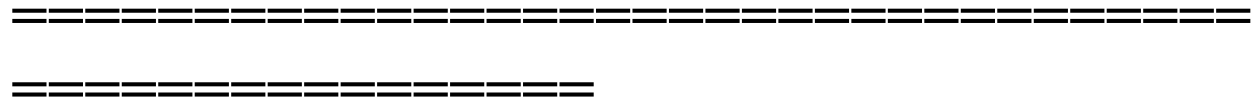
- Cheque
- Other configurable methods

Generate:

- Receipt
- Invoice
- Payment Confirmation

Allow PDF/Print.

Never expose payment data to unauthorized roles.



15. INSTALLMENT MANAGEMENT

Create installment schedules.

Fields:

- Installment Number
- Due Date
- Amount
- Paid Amount
- Pending Amount
- Status

Statuses:

- Upcoming
- Due
- Partially Paid
- Paid
- Overdue
- Waived

Create automatic overdue identification.

16. FEE REMINDERS

Create automated reminder system.

Notify students/guardians about:

- Upcoming Payment
- Payment Due
- Overdue Payment
- Partial Payment
- Receipt Generated

Notification channels should use existing LMS capabilities where available.

Keep templates editable.

17. EXAMS & ASSESSMENTS

Integrate with existing Exam system or create one if missing.

Support:

- Exam
- Test
- Quiz
- Practical
- Assignment
- Viva
- Final Assessment

Fields:

- Exam Name
- Course
- Batch
- Subject/Module

- Date
- Total Marks
- Passing Marks
- Duration
- Status

18. STUDENT RESULTS

Create result management.

Fields:

- Student
- Exam
- Subject
- Marks
- Total Marks
- Percentage
- Grade
- Result Status
- Remarks

Automatically calculate where appropriate:

- Percentage
- Grade
- Pass/Fail

Create configurable grading rules.

19. STUDENT PERFORMANCE

Create Student Performance Dashboard.

Track:

- Attendance
- Exam Results
- Assignment Completion
- Course Progress
- Practical Performance
- Trainer Feedback
- Overall Progress

Display visual analytics.

Create performance timeline.

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20. ASSIGNMENTS

Create Assignment Management.

Fields:

- Assignment
- Course
- Batch
- Module
- Student
- Trainer
- Assigned Date
- Due Date
- Submission
- Submission Date
- Marks
- Feedback
- Status

Statuses:

- Assigned
- Submitted
- Late
- Evaluated
- Pending

21. COURSE PROGRESS

Track course progress.

Display:

- Overall Progress %
- Modules Completed
- Modules Remaining
- Classes Attended
- Assignments Completed
- Tests Completed
- Practical Completed
- Certification Status

Show progress visually.

22. CERTIFICATE MANAGEMENT

Create Certificate Management.

Support:

- Course Completion Certificate
- Achievement Certificate
- Participation Certificate
- Internship Certificate
- Custom Certificate

Certificate fields:

- Certificate ID
- Student
- Course
- Issue Date
- Certificate Type

- Grade/Score
- Verification Code
- Status

Generate:

- PDF
- Print

Create certificate verification capability.

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23. STUDENT DOCUMENT MANAGEMENT

Allow secure document uploads.

Documents:

- Photo
- Identity Document
- Address Proof
- Academic Certificates
- Previous Marksheet
- Admission Form
- Agreement
- Course Documents
- Payment Documents
- Certificates
- Other Documents

Each document:

- Name
- Type
- Upload Date
- Uploaded By
- Expiry Date if applicable
- Status

Use secure access controls.

24. COMMUNICATION MANAGEMENT

Create Student Communication.

Support:

- Email
- SMS if existing provider supports it
- WhatsApp integration if available
- In-app Notifications

Send to:

- Individual Student
- Multiple Students
- Batch
- Course
- Department/Category where appropriate
- Guardians

Templates must be editable.

Maintain communication history.

25. STUDENT NOTES

Create internal notes.

Examples:

- Counselling Notes
- Academic Notes
- Behaviour Notes
- Fee Notes
- Support Notes
- Trainer Notes

- Administrative Notes

Notes must have permissions.

Allow:

- Private notes
- Role-restricted notes
- General notes

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26. STUDENT SUPPORT / ISSUES

Create Student Issue Management.

Fields:

- Issue ID
- Student
- Category
- Priority
- Description
- Assigned Staff
- Status
- Created Date
- Resolution
- Resolved Date

Statuses:

- Open
- In Progress
- Waiting
- Resolved
- Closed

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27. STUDENT STATUS MANAGEMENT

Support statuses:

- Enquiry
- Applied
- Active
- On Hold
- Completed
- Withdrawn
- Dropped
- Transferred
- Inactive

Status changes must be logged.

Do not permanently delete historical student data unnecessarily.

Use archive/soft delete where appropriate.

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28. TRANSFER MANAGEMENT

Allow student transfer:

- Batch
- Course
- Branch
- Trainer

Maintain complete transfer history:

Old Value

New Value

Reason

Date

Changed By

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29. STUDENT EXIT / WITHDRAWAL

Create Student Exit Management.

Fields:

- Exit Type
- Exit Date
- Reason
- Course Status
- Fee Clearance
- Document Clearance
- Certificate Status
- Refund
- Remarks

Exit types:

- Course Completed
- Withdrawn
- Dropped
- Transferred
- Other

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30. STUDENT BIRTHDAY

Create:

- Today's Birthdays
- Upcoming Birthdays

Allow authorized staff to send greetings.

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31. STUDENT CALENDAR

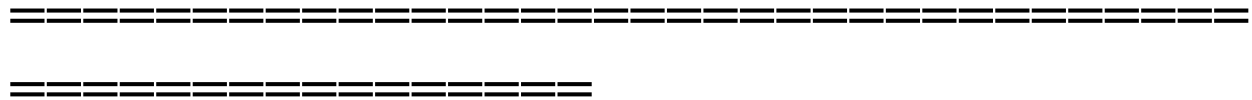
Create calendar showing:

- Classes

- Exams
- Assignments
- Fee Due Dates
- Holidays
- Student Events
- Training
- Important Dates

Support:

Day
Week
Month



32. STUDENT SEARCH

Create powerful global Student Search.

Search by:

- Student Name
- Student ID
- Admission Number
- Mobile
- Email
- Course
- Batch
- Guardian
- Payment ID

Advanced filters:

- Course
- Batch
- Branch
- Trainer
- Status
- Admission Date
- Fee Status
- Attendance
- Performance

33. STUDENT DIRECTORY

Create professional Student Directory.

Columns configurable:

- Photo
- Student ID
- Name
- Mobile
- Email
- Course
- Batch
- Trainer
- Status
- Attendance
- Fee Status
- Progress
- Admission Date

Support:

- Sorting
- Filtering
- Pagination
- Column visibility
- Bulk actions
- Export

34. BULK OPERATIONS

Allow authorized administrators to:

- Bulk assign batch
- Bulk assign course
- Bulk change status

- Bulk send notification
- Bulk import
- Bulk export
- Bulk attendance
- Bulk fee update where safe

Always confirm destructive/bulk operations.

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35. IMPORT STUDENTS

Create Excel/CSV import.

Process:

1. Upload file
2. Detect columns
3. Map columns
4. Validate records
5. Detect duplicates
6. Show errors
7. Preview
8. Confirm
9. Import
10. Show final report

Never overwrite existing data silently.

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36. EXPORT

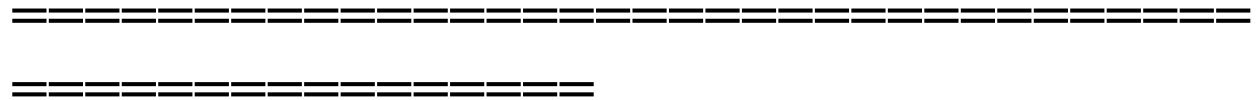
Support:

- CSV
- Excel
- PDF
- Print

Export:

- Students
- Attendance
- Fees
- Payments
- Results
- Performance
- Certificates
- Documents metadata
- Reports

Apply permission checks before export.



37. REPORTS

Create Student Reports.

Reports:

- Student Master Report
- Admission Report
- Course Enrollment Report
- Batch Report
- Attendance Report
- Low Attendance Report
- Fee Collection Report
- Pending Fee Report
- Overdue Fee Report
- Payment Report
- Exam Report
- Result Report
- Performance Report
- Assignment Report
- Course Progress Report
- Certificate Report
- Dropout Report
- Withdrawal Report
- Transfer Report
- Communication Report

Every report must support:

- Search
- Filters
- Date Range
- Sorting
- Column Selection
- Export
- Print

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38. STUDENT ANALYTICS

Create advanced analytics.

Track:

- Admission Growth
- Student Retention
- Completion Rate
- Dropout Rate
- Course Popularity
- Batch Strength
- Attendance
- Fee Collection
- Fee Outstanding
- Exam Performance
- Course Completion
- Certificate Issuance

Use real database data.

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39. ROLE BASED ACCESS

Create granular permissions.

Roles may include:

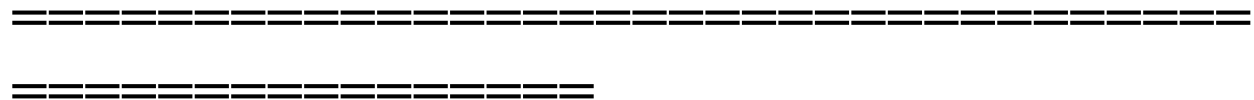
- Super Admin
- Admin
- Admission Manager
- Counsellor
- Trainer
- Accountant
- Academic Manager
- Branch Manager
- Student
- Guardian

Permissions:

- View
- Add
- Edit
- Delete
- Approve
- Export
- Print
- Manage Payments
- Manage Results
- Manage Documents
- Manage Attendance

Permissions MUST be enforced server-side.

Frontend hiding alone is NOT sufficient.



40. STUDENT LOGIN / PORTAL

If the existing LMS supports student login, integrate the Student Portal.

Student should be able to view:

- Profile
- Courses
- Batches
- Timetable
- Attendance
- Fees

- Payments
- Assignments
- Exams
- Results
- Course Progress
- Certificates
- Documents
- Notifications
- Messages

Student must NOT be able to access admin-only data.

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41. GUARDIAN PORTAL

If supported by the architecture, allow guardian access to permitted information:

- Student Profile
- Attendance
- Fees
- Payments
- Results
- Performance
- Notifications

Do not expose internal/private staff notes.

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42. NOTIFICATIONS

Create notifications for:

- Admission
- Enrollment
- Batch Assignment
- Class
- Attendance
- Low Attendance

- Fee Due
- Fee Overdue
- Payment Received
- Exam
- Result
- Assignment
- Certificate
- Course Completion
- Important Announcement

Notification templates must be editable.

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43. AUDIT LOG

Track:

- Student Created
- Student Updated
- Status Changed
- Course Changed
- Batch Changed
- Fee Changed
- Payment Added
- Payment Edited
- Result Changed
- Attendance Modified
- Document Uploaded
- Certificate Issued
- Permission Changed

Store:

- User
- Action
- Record
- Date
- Time
- Old Value
- New Value

44. SECURITY

Perform production-grade security.

Protect:

- Personal data
- Guardian data
- Payment data
- Documents
- Results
- Student records

Check:

- Authentication
- Authorization
- IDOR
- Privilege escalation
- API authorization
- File upload security
- Input validation
- XSS
- SQL injection
- CSRF
- Sensitive data exposure
- Rate limiting requirements

Never expose sensitive information unnecessarily.

Never hard-code credentials.

45. DATA INTEGRITY

Ensure:

Student

- Enrollment
- Course
- Batch
- Attendance
- Fees
- Payments
- Exams
- Results
- Assignments
- Progress
- Certificates
- Documents

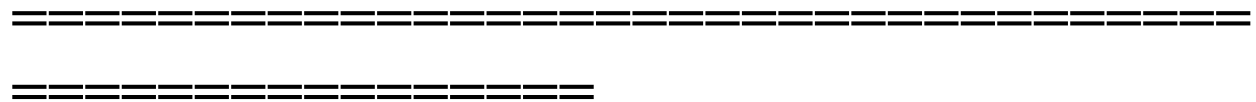
remain correctly connected.

A student may have multiple:

- Courses
- Enrollments
- Batches over time
- Payments
- Exams
- Certificates

Do NOT duplicate student master records.

Maintain historical records.



46. UI/UX

Create a premium enterprise interface.

Use:

- Responsive design
- Modern sidebar
- Top navigation
- Dashboard cards
- Charts
- Tables

- Filters
- Search
- Tabs
- Modals
- Drawers
- Status badges
- Tooltips
- Skeleton loaders
- Empty states
- Error states
- Confirmation dialogs

Support:

Light Mode

Dark Mode

Follow the existing LMS visual language.

Student profile should be visually rich but easy to navigate.

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47. FULL EDITABILITY

Avoid unnecessary hard-coded values.

Admin must be able to configure:

- Student statuses
- Admission types
- Student categories
- Course categories
- Fee types
- Payment methods
- Grading rules
- Certificate types
- Document types
- Communication templates
- Notification templates
- Branches
- Student fields
- Custom fields

48. CUSTOM FIELDS

Where practical, implement a configurable custom-field system.

Allow administrators to create:

- Text
- Number
- Date
- Dropdown
- Multi-select
- Checkbox
- File
- Long Text

Custom fields should be assignable to appropriate student records.

49. PERFORMANCE

Optimize:

- Student directory
- Search
- Filters
- Dashboard
- Reports
- Attendance
- Fees
- Payments

Use:

- Pagination
- Database indexes
- Efficient queries
- Lazy loading

- Caching where appropriate
- Debounced search

Do not load thousands of records unnecessarily.

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50. TESTING

After implementation run:

- Type checks
- Lint
- Unit tests
- Integration tests
- API tests
- Permission tests
- Database tests
- UI tests
- Browser tests

Test roles:

- Super Admin
- Admin
- Counsellor
- Trainer
- Accountant
- Academic Manager
- Student

Test workflows:

1. Create student
2. Edit student
3. Enroll course
4. Assign batch
5. Mark attendance
6. Apply leave if applicable
7. Create fee plan
8. Record payment
9. Generate receipt
10. Schedule exam

11. Enter result
12. Submit assignment
13. Track course progress
14. Generate certificate
15. Upload document
16. Send notification
17. Transfer batch
18. Withdraw student
19. Generate reports

Fix all discovered errors and retest.

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51. BROWSER QA

Open the actual running LMS in the browser.

Test the Student Management System as a real user.

Verify:

- Navigation
- Forms
- Buttons
- Search
- Filters
- Tables
- Pagination
- Charts
- Modals
- Student profile
- Attendance
- Fees
- Payments
- Exams
- Results
- Documents
- Certificates
- Reports
- Responsive UI
- Dark mode
- Permissions

If an issue is discovered:

Identify root cause

→ Fix

→ Retest

→ Check for regression

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52. DEMO DATA

If the database is empty or demo mode is required, create realistic sample data.

Include:

- Students
- Courses
- Batches
- Attendance
- Fees
- Payments
- Exams
- Results
- Assignments
- Certificates

Do NOT overwrite real existing data.

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53. FINAL INTEGRATION

The final Student Management system must integrate seamlessly with:

LMS

+

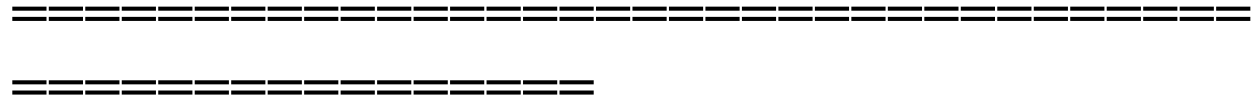
Courses

+

Batches

+

Trainers
+
Attendance
+
Fees
+
Payments
+
Exams
+
Results
+
Certificates
+
Notifications
+
Reports
+
Authentication
+
Roles & Permissions



54. FINAL QUALITY STANDARD

Do not build a simple CRUD application.

Build a complete professional Student 360° Management System.

The final experience should feel like a modern enterprise LMS/SIS platform.

Prioritize:

- Data accuracy
- Ease of use
- Attractive dashboard
- Fast search
- Complete student profile
- Automation
- Security
- Reports
- Analytics

- Role-based access
 - Historical records
 - Mobile responsiveness
 - Scalability
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55. FINAL REPORT

After implementation provide:

1. Modules completed
2. Files created
3. Files modified
4. Database tables/models created
5. Database relationships
6. APIs created
7. Permissions created
8. Reports created
9. Dashboard widgets created
10. Tests performed
11. Browser tests performed
12. Issues discovered
13. Issues fixed
14. Remaining limitations
15. Migration instructions
16. Deployment instructions

IMPORTANT FINAL RULE:

Do not fake functionality.

If an existing LMS feature already provides the required functionality, integrate with it.

If something cannot be safely implemented because of the existing architecture, identify the limitation and implement the safest compatible solution.

Do not break existing LMS functionality.

Do not delete existing student data.

Do not perform destructive database changes without a safe migration/rollback strategy.

Build the complete Student Management System inside the existing LMS.

Iska reference main attach krdunga

Ab complete analyse kro or ek final last prompt

Or jo Images upper di gyi hai un sbko combined krke proper ek prompt sare features hone chiye